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TECHNOLOGY ARCHITECTURE TECH TRENDS

Core Workout: From Technical Debt to Technical Wellness

Businesses that want to lead in the future should forgo piecemeal approaches to technical debt in favor of a new holistic frame of technical wellness

Until recently, many companies have been monitoring their technical debt—the implied cost of not modernizing systems and working with suboptimal performance—through piecemeal assessments. Instead of watching their technical debt accumulate and overwhelm their systems, however, leading organizations can establish a more continuous and complete view of their core, which can provide guidance on which technology upgrades matter and when and why to make them.

Going forward, the best proxy for understanding legacy technology realities may not be debt at all, but *health*. Organizations could treat disparate technology systems—cyber, data, infrastructure—like parts of a body, subject to thorough annual checkups. Instead of fixing aging systems that break or bottleneck IT's progress one at a time, teams can use preventative health assessments to identify and prioritize the areas of the tech stack that need treatment. Some aspects of core systems, such as the mainframes, may be in good health, only needing connectors to keep doing what they do best. Others may be due for a thorough upgrade or replacement.

Now: Yesterday's Innovations Are Aging Quickly

Whether they are ERP systems or data centers, technologies that once revolutionized business are now prone to slowing it down. As technology advances rapidly, businesses and government agencies tend to encounter two competing truths: Their future business models, existing products and services, and internal operations will be fundamentally transformed by technology, but they also struggle to invest in their infrastructure, data, applications, cyber, and workforce capabilities in a way that could adapt to that future. The result, oftentimes, can be technical debt sprawl.

To move forward, leaders can look for smarter ways to invest in modernization: a systematic assessment of their company's needs, strengths, and budget across the key areas of modernization spending. A holistic view of their organization's systems,

grounded in real business context, can help them eschew ever-growing technical debt for a more long-term view of technical health, one that improves over time and provides more confidence to business and technology teams alike.

New: Core Health Gets a Checkup

Because organizations are unclear which sources of technical debt are causing the most drag, issues can be underprioritized or poorly managed. Out of hundreds of applications or systems in a company's core, just a dozen applications that have a handful of issues may be driving the bulk of the impact of outdated systems. Instead of applying funding every year that seems to go nowhere, companies can benefit from taking a step back to tackle key issues.

A core modernization framework built around technical wellness can be more comprehensive than the traditional framing of debt. In such a framework, the focus is on preventive care—in this case, using enhanced tracking, measurement, and predictions to address suboptimal legacy technologies before they become larger issues for the business. Rather than accumulating debt and paying it off periodically with large but perhaps ineffective investments, a wellness framing would encourage businesses to iteratively pinpoint their technical ailments and predict when investments would be most effective, based on cost, operational risk, and innovation readiness.

The focus of such a wellness diagnostic, or core checkup, would be on the five areas of statistically largest spend and biggest opportunity.

Infrastructure. Infrastructure is the broadest category and often the most difficult and expensive area. Here, technical wellness translates to mainframes, servers, and end-user devices such as virtual desktops being migrated to the cloud across technical environments. In addition, aging fiber, LAN, and WAN networks across facilities such as data centers and corporate offices are on a journey to modernize to 5G, Wi-Fi 6+, low-energy Bluetooth, and satellite communications. These upgrades can enable companies to take advantage of private networks, advances in software-defined networking, and other advanced connectivity offerings.

Data. Data life cycles (including cleansing, manipulation, and management) and data storage constitute this category. Businesses should streamline data cleansing and manipulation through automation so they can spend less time on data management and more time on analyzing insights. Reporting on data usage and cleanliness, especially from a trust perspective, is also key, since AI models are only as good as the data they ingest.

Storage across data centers, offices, and remote assets can be modernized to cloud storage systems and data can even be streamed in real time.

Applications. This category includes legacy custom applications that organizations have modernized over time through one or more of the "five Rs": replatforming, revitalizing, remediating, replacing, and retrenching. It also includes package applications such as ERP and SaaS that require a clear upgrade strategy as vendors continue to update their offerings, while dealing with inevitable customizations that can complicate upgrade paths and integration.

Workforce. Many companies have a workforce comprising internal and third-party contractors who may or may not be ready to bring about modernization efforts. To improve talent acquisition and retention of tech teams, leaders should prioritize the modern engineering experience, bolstered by investments in tooling across the software development life cycle, processes, and culture.

Cyber risk and trust. Finally, companies ought to consider their relative health in cybersecurity across multiple areas: security and privacy, regulatory compliance, and ethics and morality. The first two can be tracked and improved through cybersecurity automation, especially to keep up with the growing amount of artificially generated content, while ethics requires a more nuanced approach.

Next: The Core Heals Itself

As modernization needs progress over the next decade, what if technology could become adaptive and resilient, able to "heal" its own outdated code or system with limited intervention? Consider the example of adaptive AI, which has progressed from human-initiated machine learning to unsupervised machine learning.[i] This AI can not only solve challenges but also, in studying those challenges, learns to teach and reprogram itself by developing more advanced problems.

Along these lines, core modernization solutions are also poised to become adaptive. AI embedded into core systems can currently diagnose tech debt accumulation in tech stacks and support engineers as they write the necessary code to modernize, while also streamlining remedial tasks and compliance, which often pile up when tech debt increases.[ii] These AI solutions for tech wellness are still prone to error and misdiagnoses, but as they spend more time in their residency of core systems, they're bound to improve. One day, AI could diagnose inefficiencies, develop a solution, and implement it without ever needing the support of human engineers.

As such innovations continue, longevity could be built into the five areas of core modernization from the onset. Comparable to human wellness, the goal of tech wellness would be for core systems to age gracefully, with built-in supports and checkups to allow them to fulfill their purpose.

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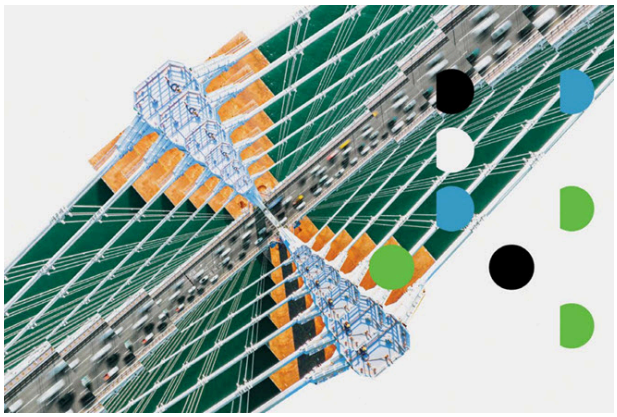

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